Priority Queues: Binary Heaps

Last Class

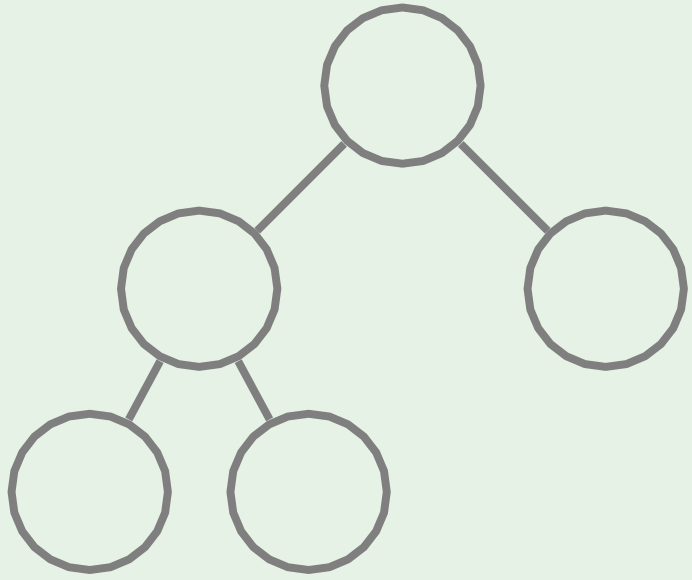
- `GetMax` works in time $O(1)$, all other operations work in time O(tree height)
- we definitely want a tree to be shallow

How to Keep a Tree Shallow?

Definition

A binary tree is complete if all its levels are filled except possibly the last one which is filled from left to right.

Example: complete binary tree



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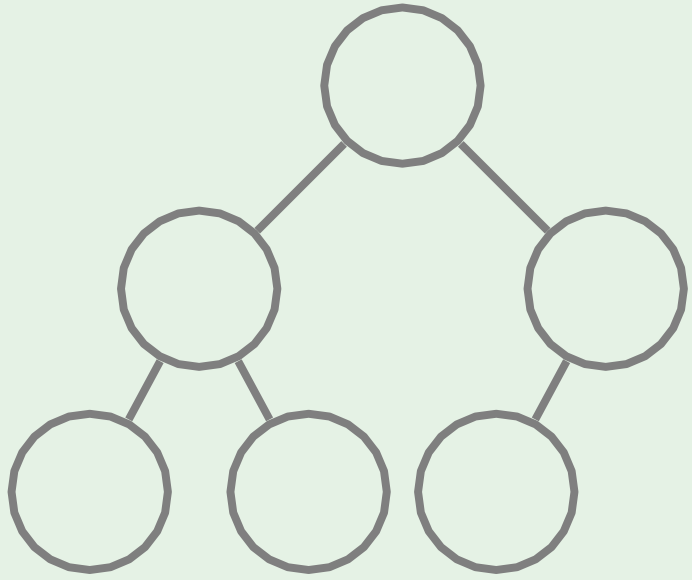

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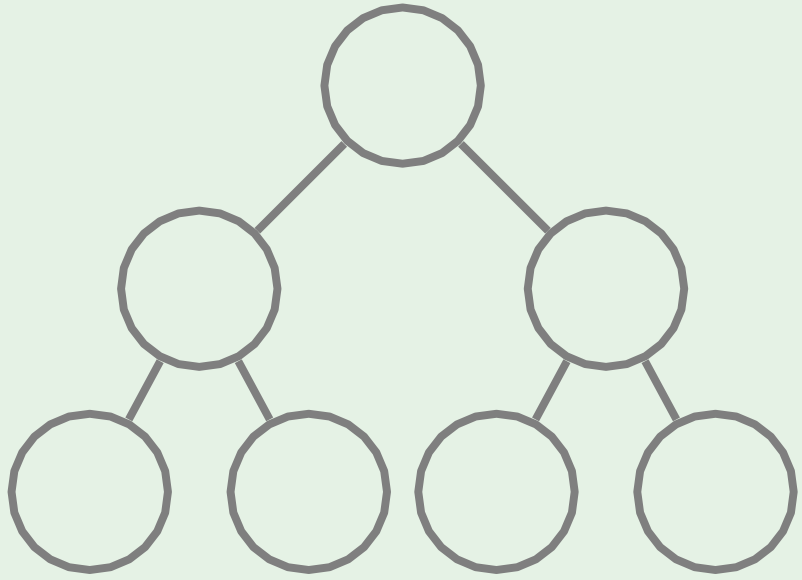

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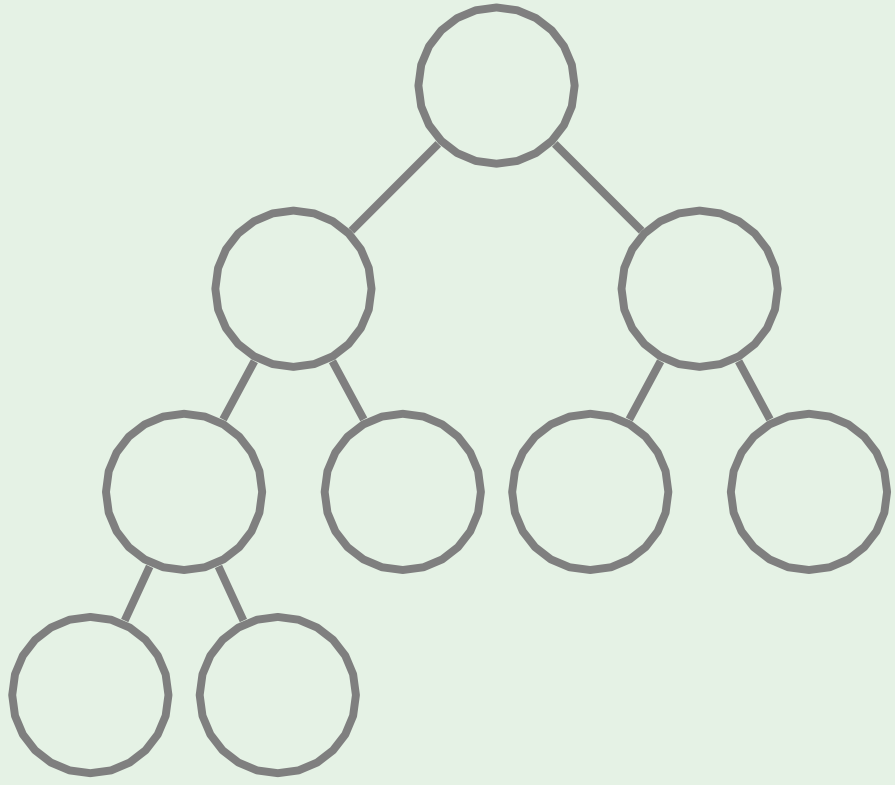
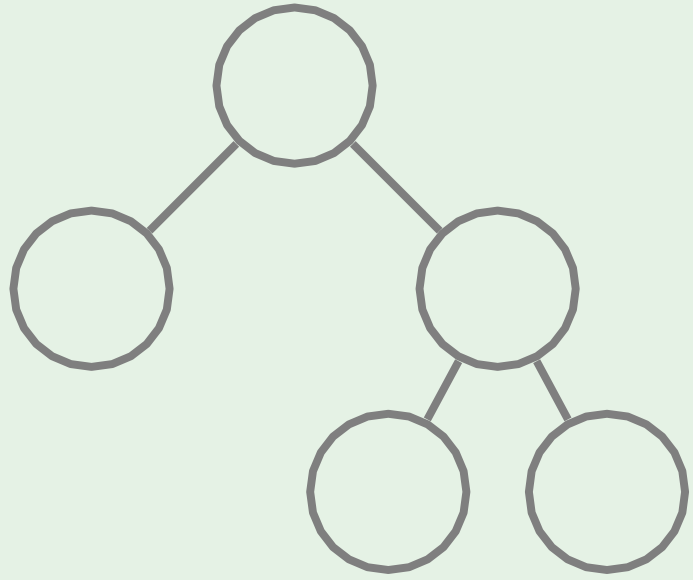


Example: not complete binary tree



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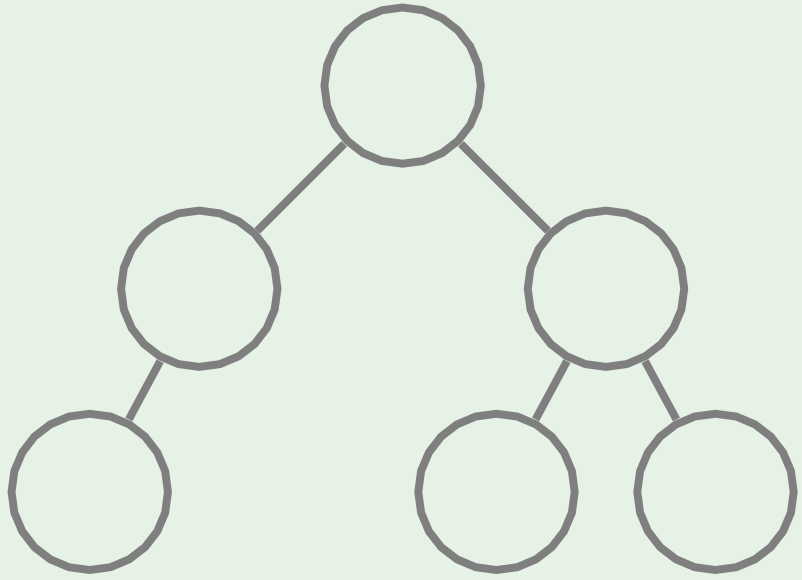

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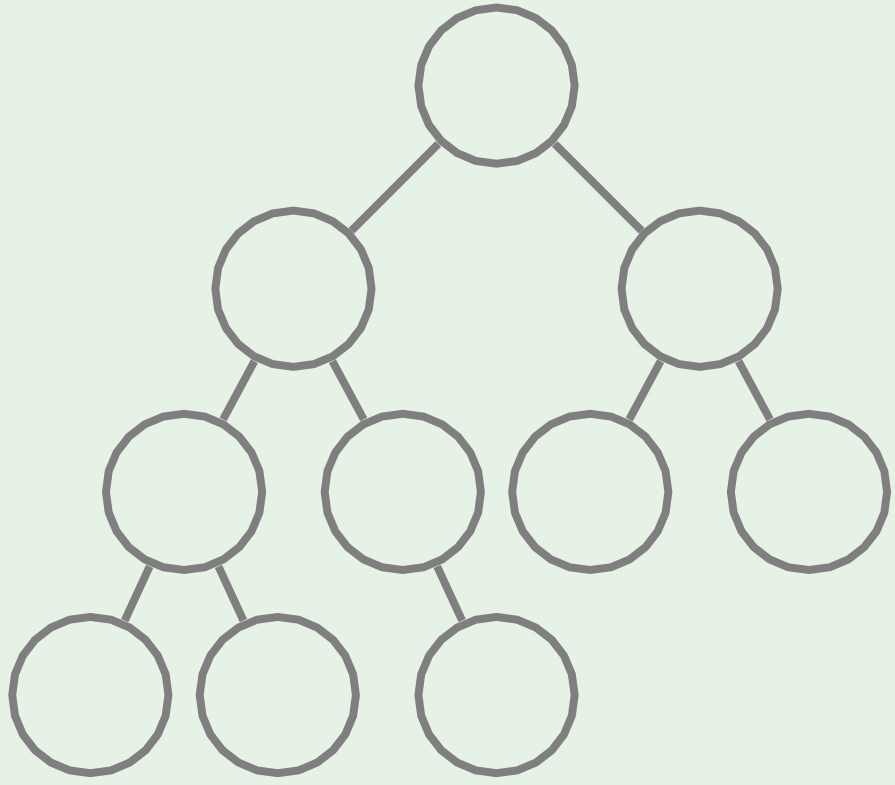

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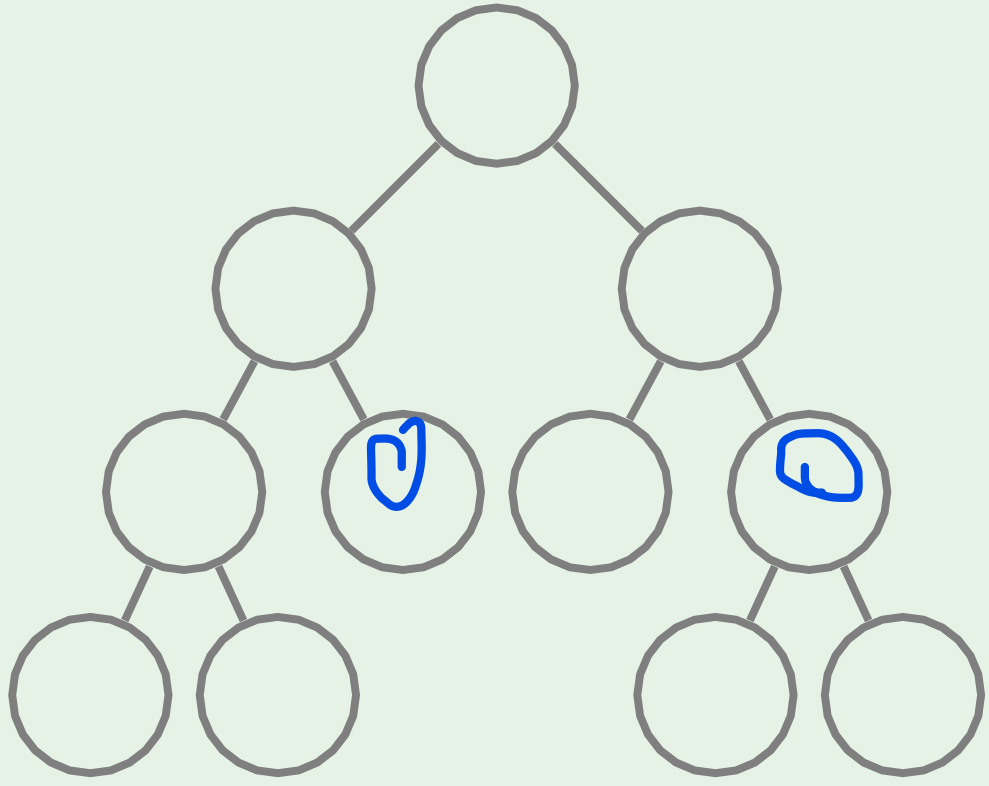


First Advantage: Low Height

Lemma

A complete binary tree with n nodes has height at most $O(\log n)$.

Proof

In a complete binary tree:

- The k-th level can hold at most 2^k nodes.
- The maximum number of nodes up to a height h (if all levels are fully filled) is given by the sum of a geometric series:

$$1 + 2 + 4 + \cdots + 2^h = 2^{h+1} - 1$$

- So, a complete binary tree with height h has at most $2^{h+1} - 1$ nodes.

Let n be the total number of nodes in the complete binary tree. Then:

$$n \leq 2^{h+1} - 1$$

Solving for h, we get:

$$2^{h+1} \leq n + 1$$

Taking the logarithm on both sides:

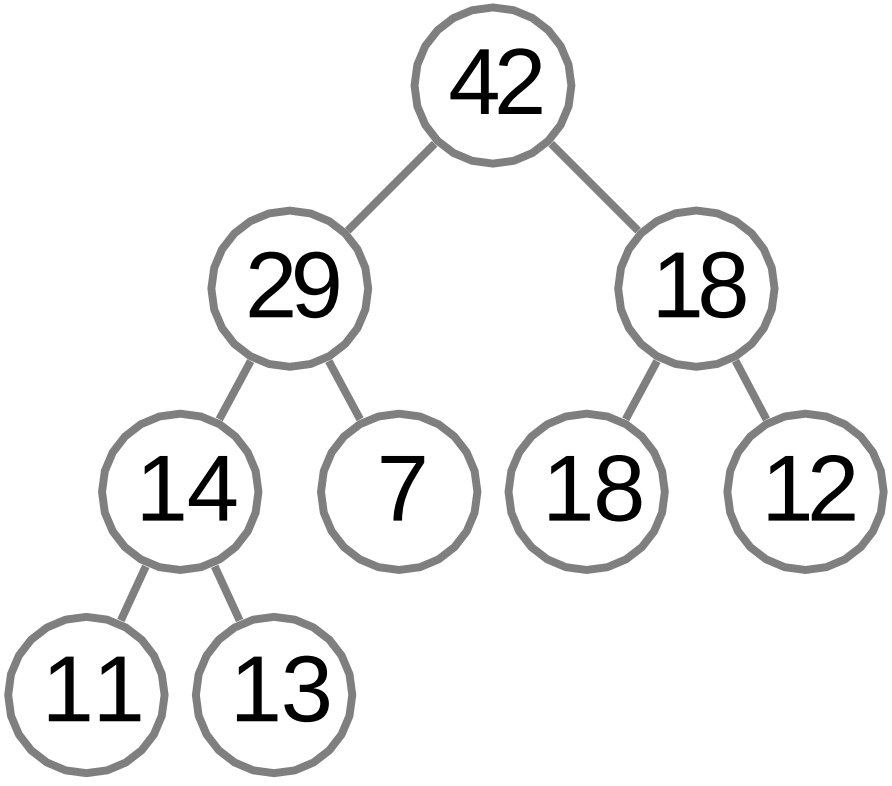
$$h + 1 \leq \log_2(n + 1)$$

So:

$$h \leq \log_2(n + 1) - 1$$

This shows that the height h is at most $O(\log n)$.

Second Advantage: Store as Array



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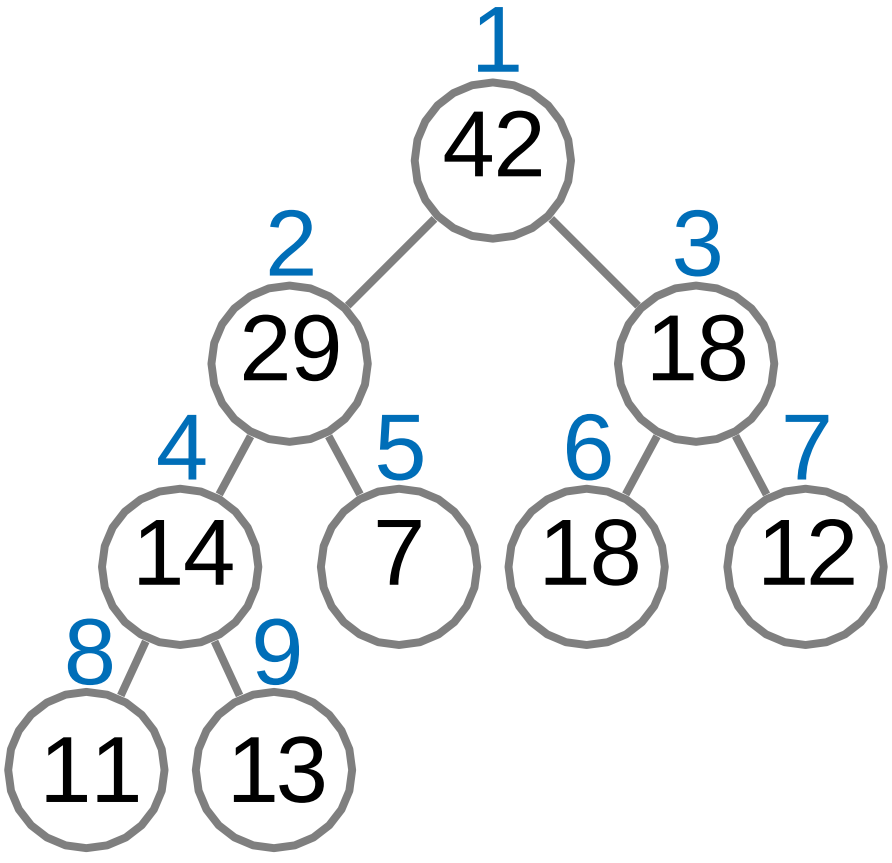

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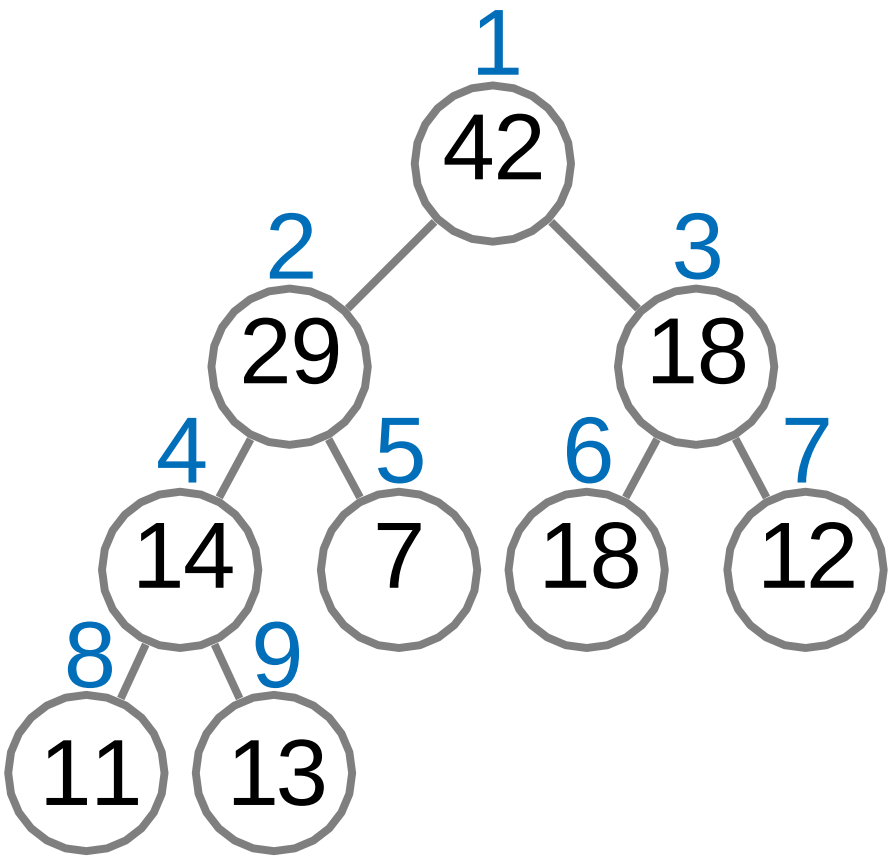

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leftchild(i) = $2i$

rightchild(i) = $2i + 1$

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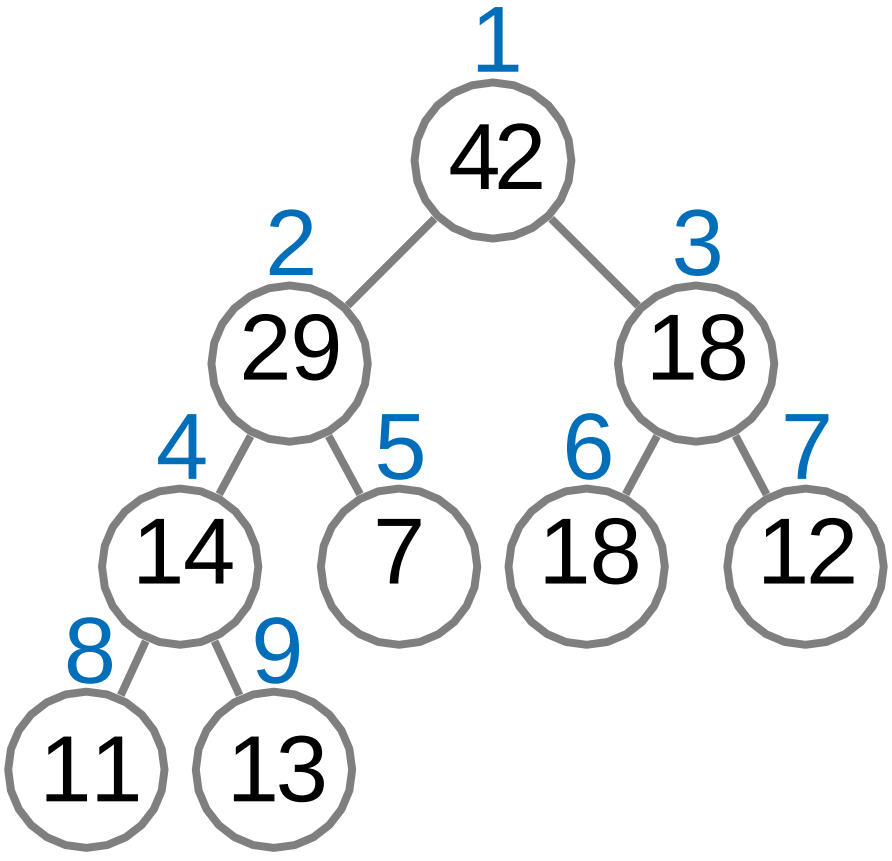

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1	2	3	4	5	6	7	8	9
42	29	18	14	7	18	12	11	13

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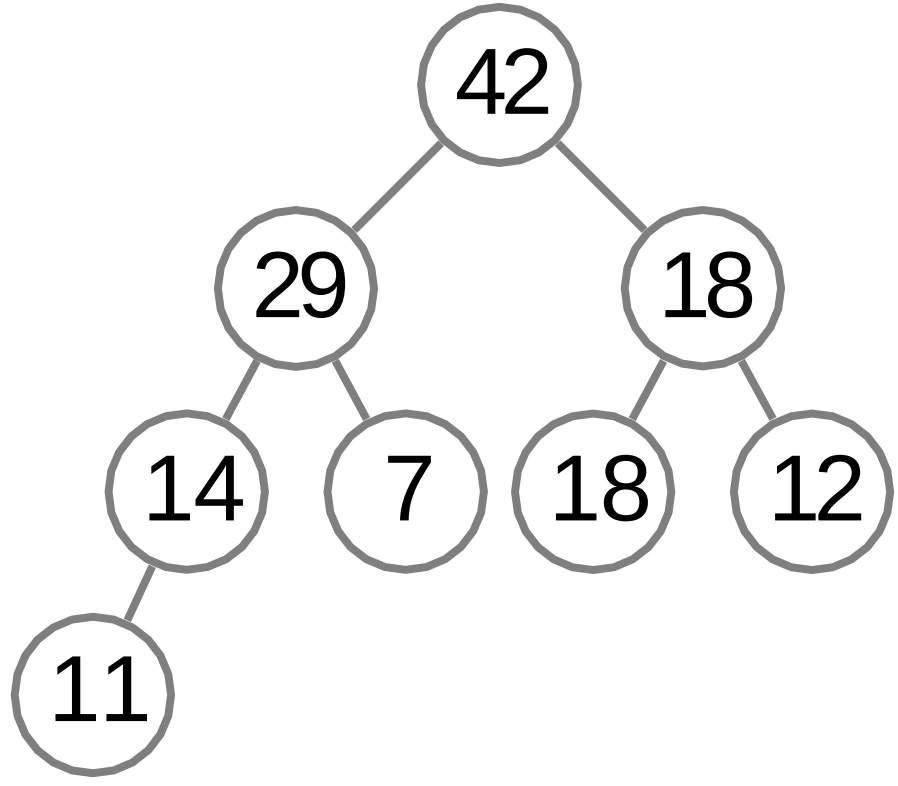
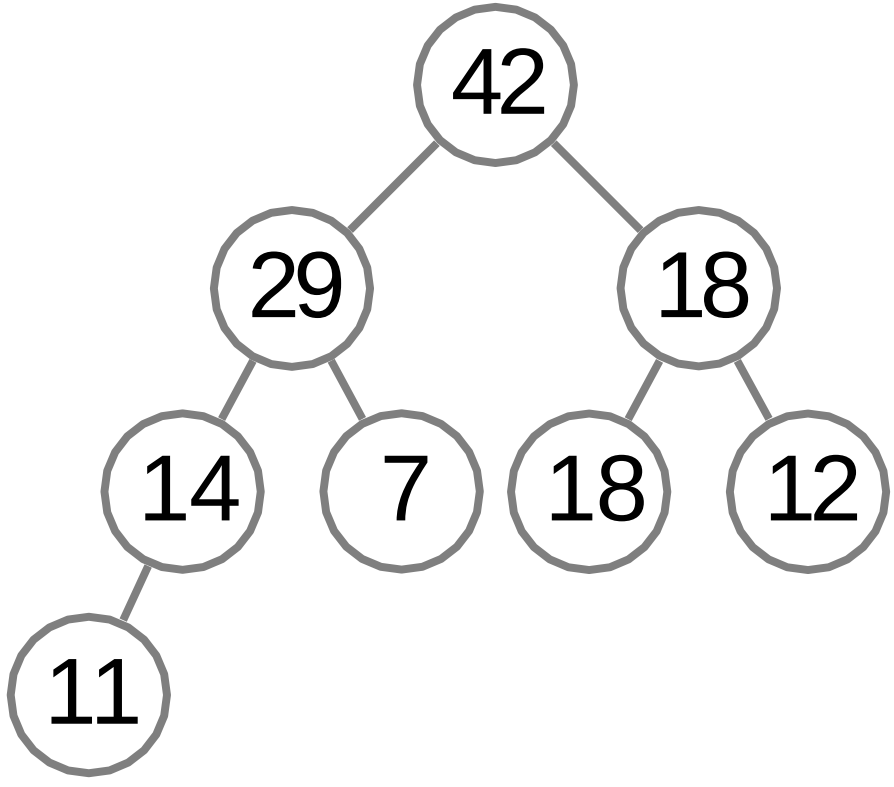
- We need to keep the tree complete.
- Which binary heap operations modify the shape of the tree?
- Only `Insert` and `ExtractMax` (`Remove` changes the shape by calling `ExtractMax`).

Keeping the Tree Complete



Keeping the Tree Complete

to insert an element, insert it as a leaf in the leftmost vacant position in the last level and let it sift up



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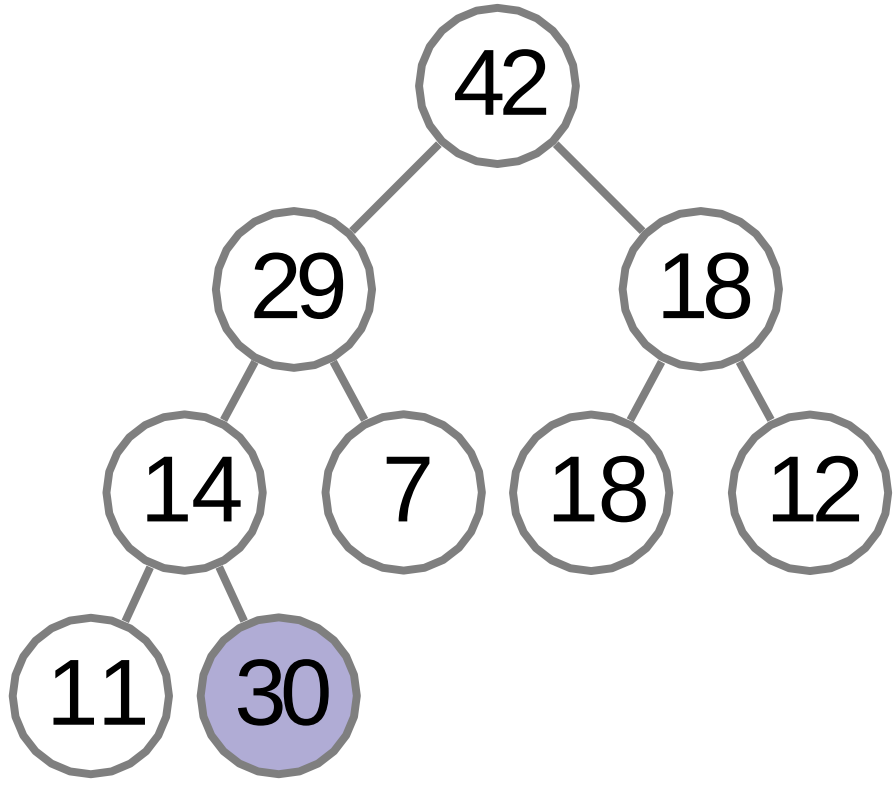

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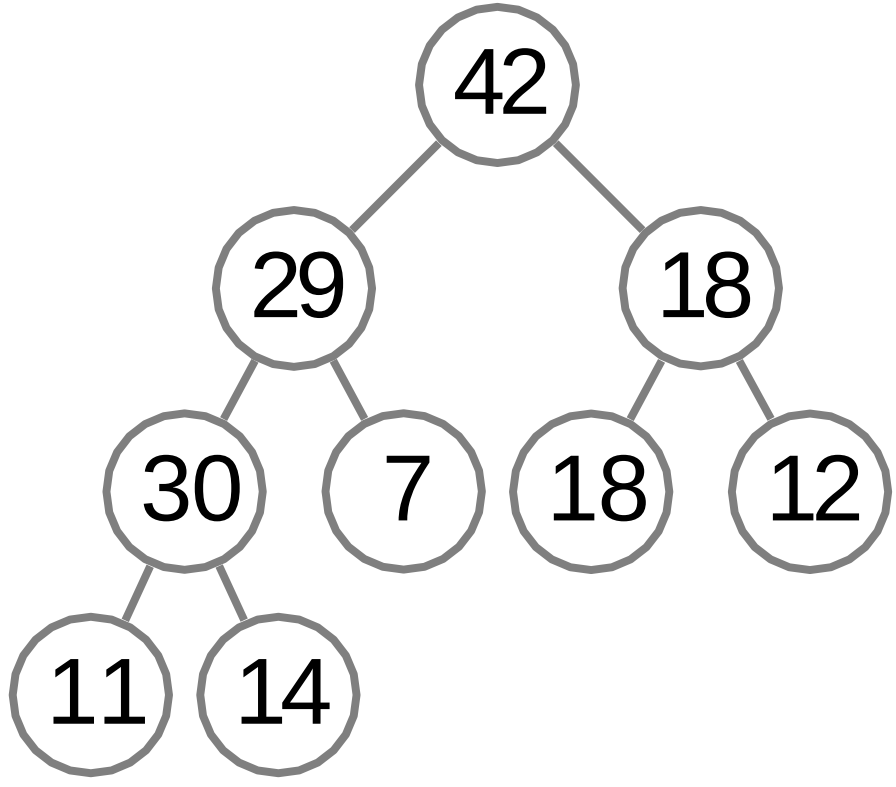

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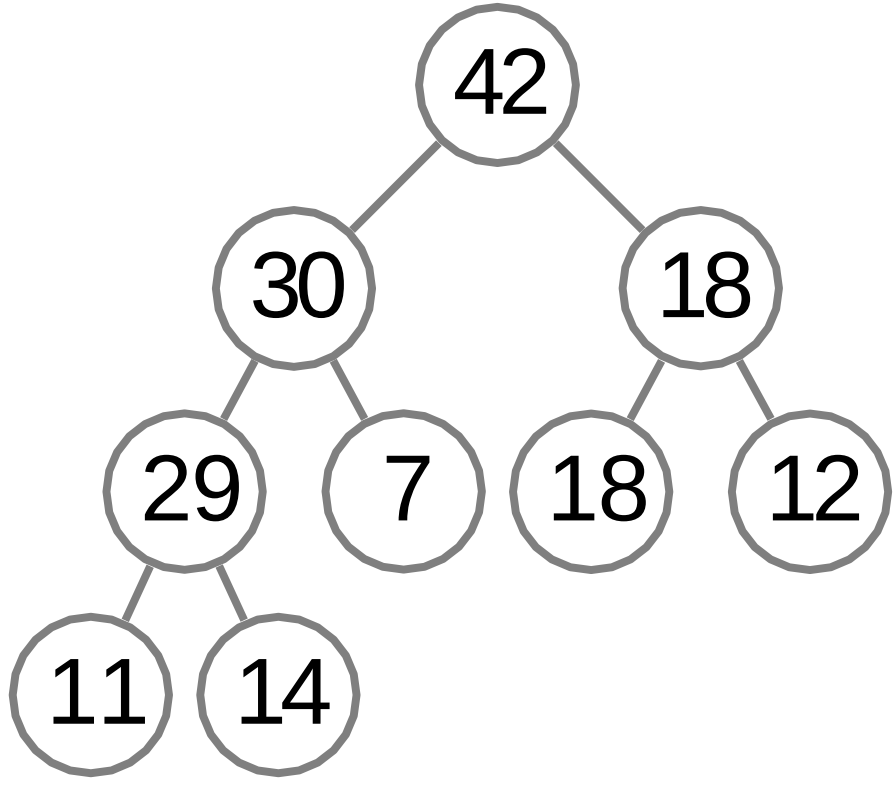
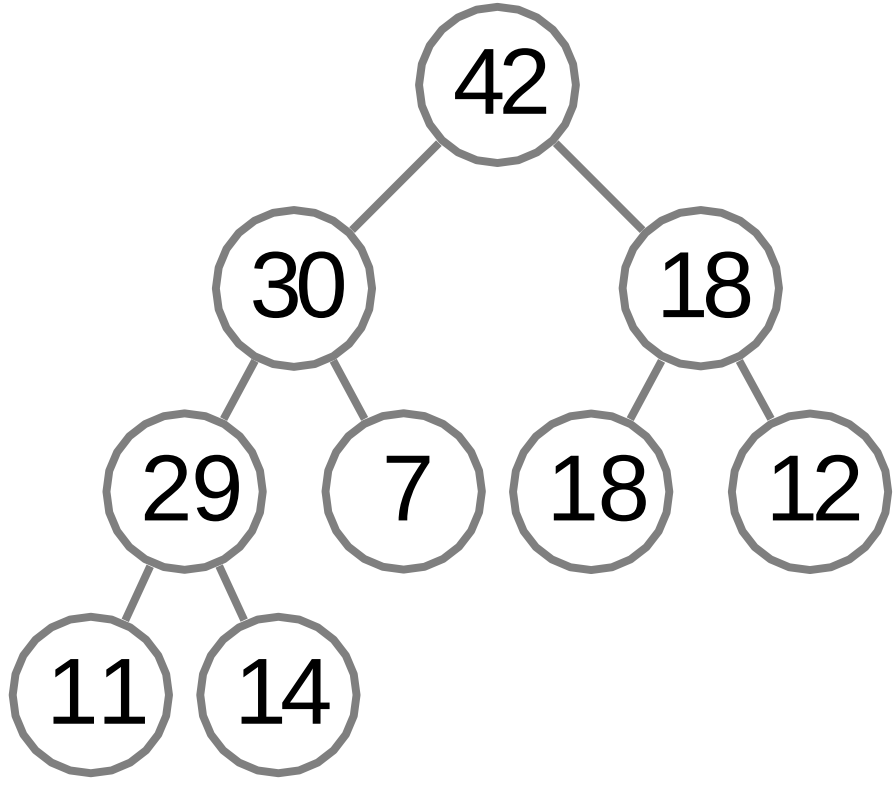


Keeping the Tree Complete

to extract the maximum value, replace the root by the last leaf and let it sift down



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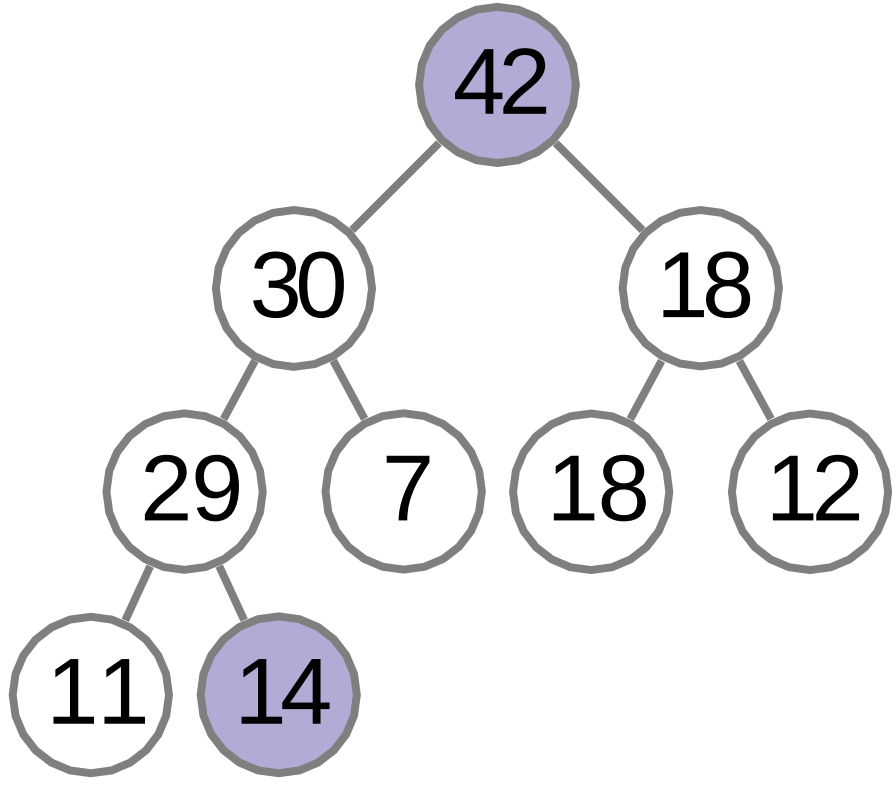

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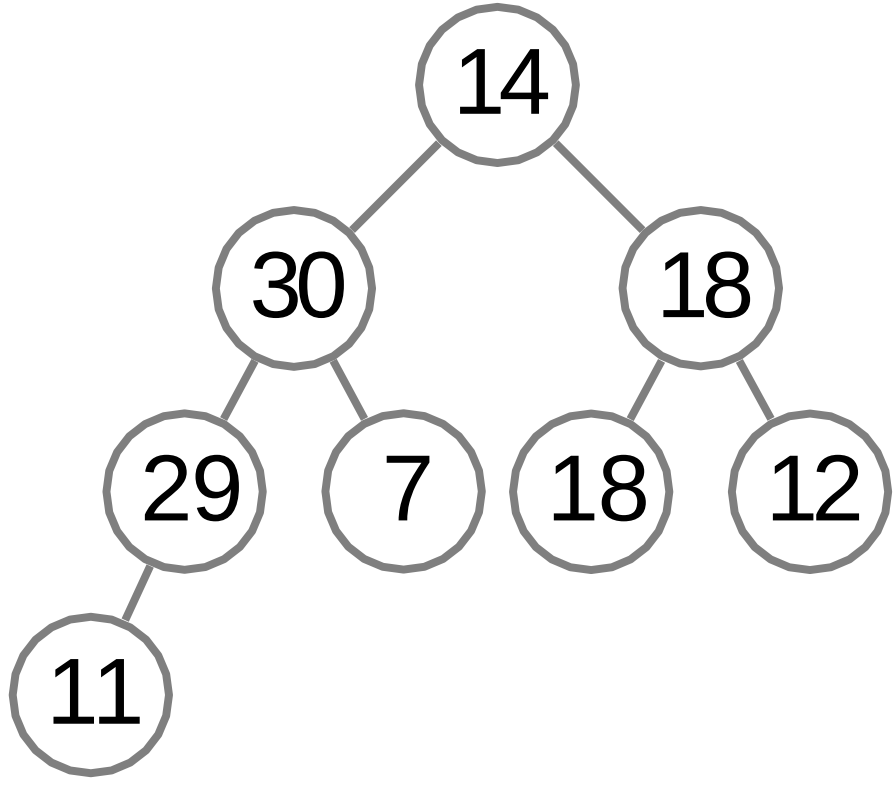

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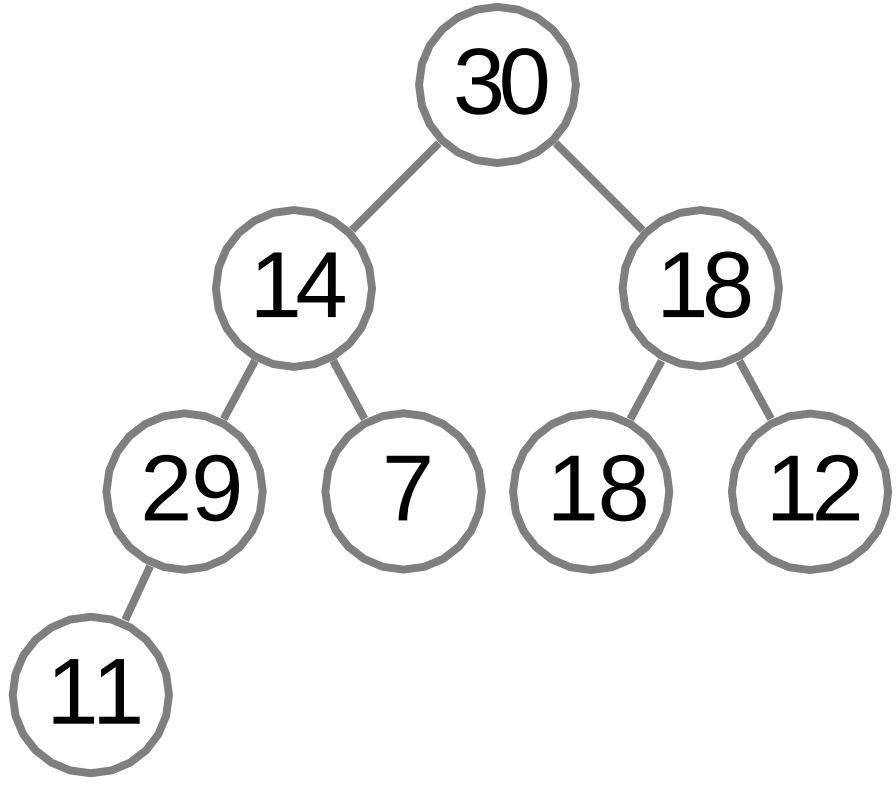

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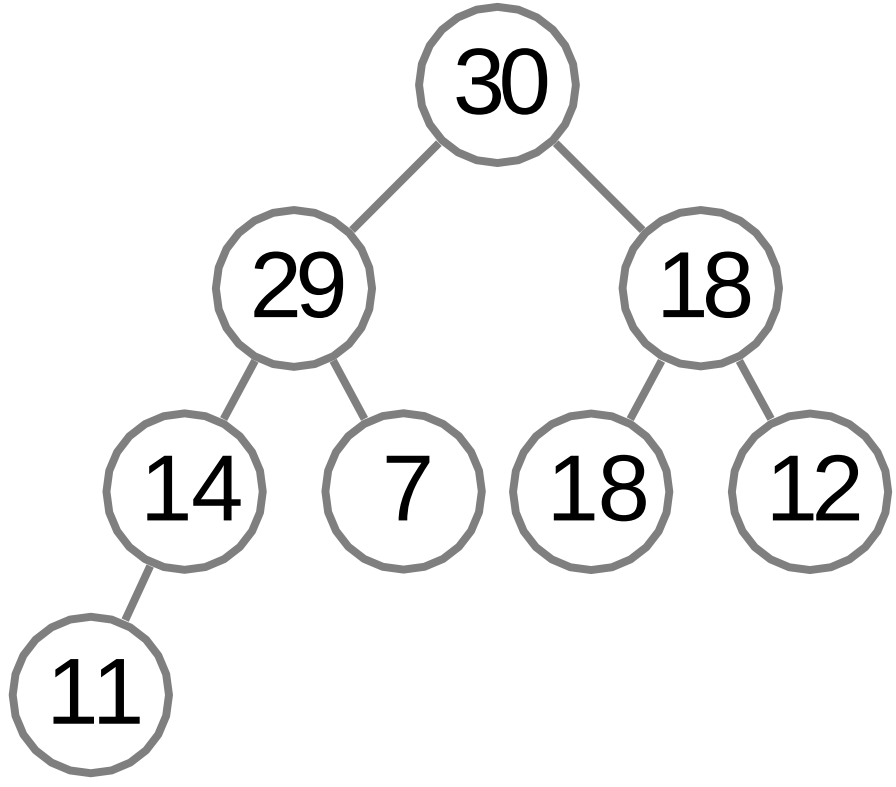


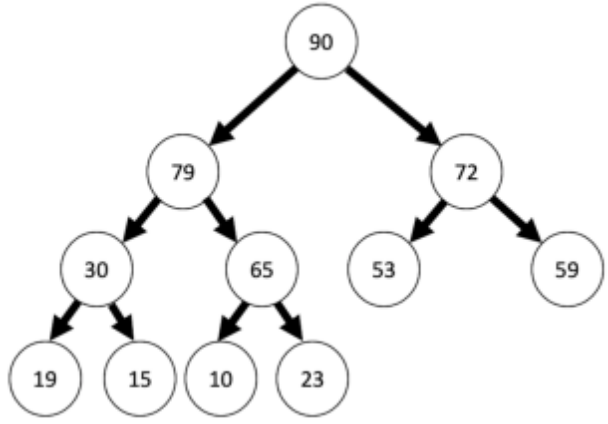
Question 1

(a) Show that the complete binary tree with n nodes has height at most O(log n).
(b)If a complete binary tree has 127 nodes, calculate the maximum possible height of the tree.

Question 2

(a) Consider the following max heap represented as a binary tree:

Show the process of removing the node with value 23 from the heap. Draw all intermediate trees after each step of rebalancing the heap.



(b) Given that a complete binary tree has a height of O(log n), justify why the time complexity of removing a node from a heap and restoring the heap property is O(log n).